About NineScrolls LLC

LEADING INNOVATION IN SCIENTIFIC RESEARCH EQUIPMENT

NineScrolls LLC is a dynamic start-up company dedicated to advancing innovation and integration in the scientific research equipment industry. Our primary focus is on establishing a comprehensive platform that connects manufacturers, researchers, and industry professionals across the United States.

By fostering collaboration and streamlining access to cutting-edge laboratory equipment, we aim to empower scientific discovery and drive technological advancements. At NineScrolls LLC, we are committed to delivering tailored solutions and creating value for our partners and clients through expertise, efficiency, and innovation.

Integration

We create seamless connections between manufacturers, researchers, and industry professionals to advance scientific discovery.

Innovation

We drive advancement in the scientific equipment industry through innovative solutions and platforms.

Collaboration

We foster partnerships and facilitate connections across the scientific community to accelerate progress.

Expertise

We leverage deep industry knowledge to deliver tailored solutions that create value for our partners and clients.

Peer-Reviewed Validation for the Platforms We Represent

Research using corresponding plasma, deposition, and vacuum process platforms has appeared in Nature Portfolio journals, Advanced Materials, Materials Today, and Scientific Reports.

Real published research using corresponding process platforms.

Nature Communications · 2021
Near-ideal van der Waals rectifiers based on all-two-dimensional Schottky junctions
Corresponding RIE process platform · 245 citations (as of Jul 2026)

Light: Science & Applications · 2026
On-chip nonlocal metasurface for color router
Corresponding RIE process platform

Advanced Materials · 2026
Diffraction-free omnidirectional antireflection binary metasurface
Corresponding ICP process platform

Materials Today · 2026
Solar-blind deep-UV photodetector based on β-Ga_2O_3/AlN/p-Si
Corresponding PECVD process platform · 9 citations (as of Jul 2026)

Scientific Reports · 2025
Experimental study of inductively coupled plasma etching of patterned single crystal diamonds
Corresponding ICP process platform · 4 citations (as of Jul 2026)

These publications validate represented platform classes and process capabilities. They are not claims of NineScrolls-branded installed-base citations.

EQUIPMENT PLATFORM

RIE Etcher Series

Uni-body Design Concept Foot-print outstanding (ref 1.0m × 1.0m).

Uniform Chamber Center Pump-down Better process performance.

Showerhead Gas Feed-in Tuned as a preset parameter dependently.

Configurable Plasma Discharge Gap Tuned as a preset parameter dependently.

Cost or Performance Orientation RF, pump, valves etc. depending on requirements.

Sample Handling Options Open-Load or Load-Lock.

Specification	Parameters
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Etching Materials	Si-Based (Si/SiO2/SiNx/SiC/Quartz etc.), Compounds (InP/GaN/GaAs/Ga2O3/ZnS etc.), 1D&2D Materials (MoS2/BN/Graphene etc.), Metals (Au/Pt/W/Ta/Mo etc.), Failure Analysis, etc.
Vacuum	TMP & Mechanical Pump
RF Power	Full Range 300-1000W, optional
Gas System	4 lines (Standard) or customized
Wafer Cooling	Water Cooling or He Backside Cooling optional
Wafer Stage Temperature Range	From -70°C to 200°C, optional
Non-Uniformity	Less than ±5% (Edge Exclusion)

EQUIPMENT PLATFORM

ICP Etcher Series

Uni-body Design Concept Foot-print outstanding (ref 1.0m × 1.5m).

Process Design Kits Better process performance.

Chamber Control Chamber liner, electrode temperature control suitable for different process application.

Configurable Plasma Discharge Gap Tuned as a parameter dependently.

Cost or Performance Orientation RF, pump, valves etc. depending on requirements.

Plasma Specialization Low power plasma technology, ion damage-free optional.

Sample Handling Options Open-Load or Load-Lock.

Specification	Parameters
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Etching Materials	Si-Based (Si/SiO_2/SiN_x/SiC/Quartz etc.), Compounds (InP/GaN/GaAs/Ga_2O_3 etc.), 2D Materials (MoS_2/BN/Graphene etc.), Metals (W/Ta/Mo etc.), Diamond, Failure Analysis, etc.
Vacuum	TMP & Mechanical Pump
RF Power	Source 1000-3000W, Bias 300-1000W, optional
Gas System	5 lines (Standard) and He backside cooling, or customized
Wafer Stage Temperature Range	From -70°C to 200°C, optional
Non-Uniformity	Less than ±5% (Edge Exclusion)

EQUIPMENT PLATFORM

Plasma Photoresist Stripping Series

Uni-body Design Concept Foot-print outstanding (ref 0.8m × 0.8m).

Uniform Chamber Center Pump-down Better process performance.

Uniform Gas Feed-in Tuned as a preset parameter dependently.

Configurable Plasma Discharge Gap Tuned as a preset parameter dependently.

Cost or Performance Orientation RF, pump, valves etc. depending on requirements.

Sample Handling Open-Load.

Specification	Parameters
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Etching Materials	Organics (PR/PMMA/PS nanosphere etc.), 2D Materials (MoS_2/BN/Graphene etc.), Failure Analysis, etc.
Vacuum	Mechanical pump
RF Power	Full range 300-1000W, optional
Gas System	2 lines (Standard) or customized
Wafer Cooling	Water cooling
Wafer Stage Temperature Range	From 5°C to 200°C, optional
Non-Uniformity	Less than ±5% (Edge Exclusion)

EQUIPMENT PLATFORM

IBE/RIBE Series

Uni-body Design Concept Foot-print outstanding (ref 1.0m × 0.8m).

Maintenance and Sample-handling Friendly Sample holder and ion source design for easy-to-use operation.

Flexible Ion Source Design Different kinds of ion source easy-to-swap design, depending on customer requirements.

Cost or Performance Orientation Ion source, pump, valves etc. depending on requirements.

Sample Handling Options Open-Load or Load-Lock.

Specification	Kaufman ion source	RF ion Source
Wafer Size Range	up to 6 inch	up to 12 inch
Gas System	1 line (standard) or customized	3 line (standard) or customized
Wafer Stage Motion	Tilt from 0° to 90°, Rotation from 1-10 rpm/min	
Wafer Stage Cooling	From 5 to 20°C, Water cooling; He backside cooling optional	
Vacuum	TMP & Mechanical Pump	
Base Vacuum	Better than 7E-7 Torr	
Non-Uniformity	Less than ±5% (Edge Exclusion)	

EQUIPMENT PLATFORM

ALD Series

Uni-body Design Concept Foot-print outstanding (ref 0.8m × 1.0m).

Box-in-Box Process Chamber Better process performance.

Configurable Gas Feed-in Showerhead gas feed-in, tuned as a preset parameter independently.

High-AR Step Coverage Excellent high-AR step covering capability with multiple gas inlets and vertical precursor throw.

Cost or Performance Orientation RF, pump, valves etc. depending on requirements.

Sample Handling Options Open-Load or Load-Lock.

Specification	Parameters
Wafer Size Range	4, 6, 8, 12 inch or Supersize optional
Growth Materials	Oxides (Al_2O_3/HfO_2/SiO_2/TiO_2/Ga_2O_3/ZnO etc.), Nitrides (TiN/TaN/SiN_x/AlN/GaN etc.), Metals (Pt/Pd/W etc.), etc.
Vacuum	TMP & Mechanical Pump
Base Vacuum	Better than 5E-5 Torr
RF Power	Remote Plasma 300-1000W, optional
Number of Precursor	2-6 lines or customized
Temperature of Source	From 20°C to 150°C (Standard), 200°C optional
Wafer Temperature Range	From 20°C to 400°C, higher temperature optional
Non-Uniformity	Less than ±1% (Al_2O_3)

EQUIPMENT PLATFORM

PECVD Series

Uni-body Design Concept Foot-print outstanding (ref 1.0m × 1.0m).

Process Design Kits Better process performance.

Variable Plasma Discharge Gap Better process performance.

Temperature Control Chamber liner, electrode temperature control suitable for different process application.

Advanced RF System Electrode RF driven (13.56MHz and/or 400KHz) for better process tuning and control, low stress.

Cost or Performance Orientation RF, pump, valves etc. depending on requirements.

Sample Handling Options Open-Load or Load-Lock.

Specification	Parameters
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Deposition Materials	Si-Based (α-Si:H/SiO2/SiNx/SiC etc.), etc.
Vacuum	Roots & Mechanical Pump
RF Power	Full Range 500-2000W, optional
Gas System	6 lines (Standard) or customized
Wafer Stage Temperature Range	From 20°C to 400°C, higher temperature optional
Non-Uniformity	Less than ±5% (Edge Exclusion)

EQUIPMENT PLATFORM

HDP-CVD Series

Uni-body Design Concept Foot-print outstanding (ref 1.0m × 1.5m).

Process Design Kits Better process performance.

Temperature Control Chamber liner, electrode temperature control suitable for different process application.

Step Coverage Excellent step covering capability, tuned as a parameter dependently.

Cost or Performance Orientation RF, pump, valves etc. depending on requirements.

Sample Handling Options Open-Load or Load-Lock.

Specification	Parameters
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Deposition Materials	Si/SiO2/SiNx/SiON/SiC, etc.
Vacuum	TMP & Mechanical Pump
RF Power	Full Range: Source 1000-3000W, Bias 300-1000W, optional
Gas System	6 lines (Standard) or customized
Wafer Stage Temperature Range	From 20°C to 200°C
Non-Uniformity	Less than ±5% (Edge Exclusion)

EQUIPMENT PLATFORM

PVD Magnetron Sputtering Series

Uni-body Design Concept Foot-print outstanding (ref 1.0m × 1.7m).

Creative Magnetron Design Magnetron target structure self-designed, designed and modified based on customer needs.

Flexible Target Configuration Magnetron target face-down or face-up optional, angle tiltable and deposition distance tunable.

Advanced Electrode Control Electrode rotational and temperature controllable, suitable for different process application.

RF Bias Capability Substrate can be RF biased for in-situ clean, and better process tuning and control.

Cost or Performance Orientation RF, pump, valves etc. depending on requirements.

Sample Handling Options Open-Load or Load-Lock.

Specification	Parameters
Wafer Size Range	4, 6, 8, 12 inch or multi-wafers optional
Magnetron Sputtering Source	2-6 optional
Substrate Temperature	Water-cooling, 400°C, 800°C, 1200°C, optional
Gas System	2 lines (Standard), number of lines customized
Power	DC or RF customized, automatic switcher
Non-Uniformity	Less than ±5%
Pre-Cleaning	Independent chamber or in-situ, RF plasma, optional
Base Pressure	Better than 5E-7 Torr, higher vacuum customized

EQUIPMENT PLATFORM

Coater/Developer & Hotplate Series

Uni-body Design Concept Foot-print outstanding (ref 1.0m × 0.8m).

Flexible Configuration Number of coater/developer/hotplate modules customized.

Modular Options Wide range of options down to module level, including dispense systems, temperature for developers etc.

Cost or Performance Orientation Dispense, pump, valves etc. depending on requirements.

Sample Handling Open-Load.

Specification	Coater	Developer
Wafer Size Range	Small-piece, 2, 4, 6, 8, 12 inch or Square optional	
Max. Spin Speed	8000 rpm ±1rpm	5000 rpm ±1rpm
Max. Acceleration	8000 rpm/s	5000 rpm/s
Dispense Arm	Up to 2 photoresist lines	Up to 2 developer lines and deionized water line
Interlock	Vacuum pressure, uncover etc.	

Hotplate Specifications	
Wafer Size Range	Small-piece, 2, 4, 6, 8, 12 inch or Square optional
Max. Temperature	Up to 200°C, Higher Temperature optional
Lift-Pins	3 lift-pins, minimum compatible 2 inch

EQUIPMENT PLATFORM

Plasma Cleaner Systems

Ultra-Compact Footprint One-piece integrated design for space-limited laboratories (ref 630 mm × 600 mm).

Maintenance and Sample-handling Friendly Simple chamber structure with easy access; designed for fast loading, cleaning, and routine maintenance.

Stable and Cost-Effective Performance Optimized RF plasma design for repeatable surface treatment; excellent cost-performance ratio for research and light manufacturing.

Flexible Process Capability Supports surface cleaning, activation, and modification; single-wafer or multi-wafer batch processing.

Multi-Gas Plasma Processing Compatible with O_2 / N_2 / Ar plasma processes; supports hydrophilic / hydrophobic surface treatments.

Tabletop / Bench-top Design Single or multi-wafer batch processing; compatible with 6-inch and smaller wafers.

Specification	Parameters
Wafer Size Range	≤ 6 inch, multi-wafer batch processing
RF Power	0 ~ 300 W / 500 W, automatic matching
Gas System	2 ~ 3 gas lines
Process Gases	O_2, N_2, Ar
Flow Control Range	0 ~ 300 sccm
Flow Control	MFC or manual control
Pump System	Mechanical pump (TMP optional)
Operation	Touchscreen control, fully automated
Footprint	630 mm × 600 mm
Compatible Materials	Photoresist (PR); PMMA; PDMS; HMDS; organic films and polymers; semiconductor materials; optical materials; biomedical materials
Main Functions	Surface cleaning; surface activation; hydrophilic / hydrophobic treatment; functional group modification (–OH / –H / –COOH); contact-free plasma processing
Typical Applications	Chemical & biological laboratories; failure analysis; optical components; biomedical and medical devices

Family options: HY-4L · HY-20L · HY-20LRF · PLUTO-T · PLUTO-M · PLUTO-F

EQUIPMENT PLATFORM

E-Beam Evaporation Series

Multi-Source E-Beam and Thermal E-beam and thermal resistance sources for metals, oxides, fluorides, and IR films.

In-situ Endpoint Control In-situ QCM thickness monitoring and endpoint control.

Optical and IR Stack Ready High-purity films for photonic crystals, optical multilayers, and IR sensors.

Flexible Operating Modes Manual, semi-automatic, or full-automatic operation.

Directional Lift-off Deposition Directional deposition suited to lift-off metallization and patterning.

Specification	Parameters
Substrate	1×8 in or 5×4 in
E-Gun Crucible	6 pockets, 17 cc each
Uniformity	±3-5%
Thickness Control	In-situ QCM endpoint
Vacuum	~8×10^{-4} Pa
Operating Modes	Manual / semi-auto / full-auto
Sources	E-beam + thermal resistance
Materials	Metals, oxides, fluorides, IR films

Office Location

12546 Cabezon Pl
San Diego, CA 92129
United States

Business Hours

Monday – Friday
9:00 AM – 5:00 PM PST

Contact Information

General Inquiries: info@ninescrolls.com

Sales: sales@ninescrolls.com

Urgent Matters: +1 (858) 879-8898

Technical Support

Support Email: support@ninescrolls.com

Support Hours: Monday – Friday, 8:00 AM – 6:00 PM PST

Emergency Support: 24/7 available for critical issues